

GE261004 DESIGN THINKING

Unit I – Introduction to Design Thinking · Presentation 1 of 6

Design Thinking: Concept and Importance

What design thinking is · where the idea came from · well-defined and wicked problems · why it matters to engineers · the mindsets it asks for

Dr. Anantha Raman L., M.E., Ph.D. · Associate Professor, Department of Mechanical Engineering, Rajalakshmi Institute of Technology, Chennai

In this presentation

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- Where the idea came from
- Two kinds of problems
- What design thinking is not
- Why it matters to engineers
- Where design thinking is used
- The mindsets of a design thinker
- What this course will ask of you

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What design thinking is

A definition, its origins, and the kind of problem it is built for

Design thinking – a working definition

- A human-centred approach to problem solving that starts with the people who will live with the solution, not with the technology available
- **A process** you can follow stage by stage, and a **mindset** – habits of thought you bring to any problem
- A skill that is learnt and practised; it is not a talent reserved for designers or artists
- Its output is not a drawing – it is a tested solution that people actually want

KEY POINT Design thinking does not begin with “what can we build?” It begins with “whose problem is this, and what is it really?”

Where the idea came from

- **1969** – Herbert Simon: designing is a way of thinking in its own right
- **1970s–80s** – Nigel Cross shows designers reason unlike scientists
- **1991** – IDEO makes user observation and rapid prototyping a profession
- **2005** – the Stanford d.school teaches it to every discipline
- **2008** – Tim Brown brings it into mainstream management

KEY POINT The way of working is old. What is new is teaching it as a repeatable process to non-designers – including engineers.

Two kinds of problems

Well-defined problem

- Requirements are given
- One correct answer exists
- Success is easy to measure
- Solve it once, it stays solved

“Calculate the beam deflection”

Ill-defined (wicked) problem

- The problem must be found first
- Many answers, none 'correct'
- Success depends on whose view
- Each answer reshapes the problem

“Make commuting safer for women”

Your courses train you on the left. Real projects arrive from the right.

What design thinking is – and is not

Design thinking is not

- Decoration or styling added at the end
- Guesswork dressed up as creativity
- A replacement for engineering analysis
- One brainstorm on one afternoon

Design thinking is

- A discipline for finding the right problem
- Evidence gathered from real users
- A partner to analysis – it decides what to analyse
- A cycle you run again and again

KEY POINT Analysis tells you whether a solution works. Design thinking tells you whether it is the right solution to build.

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Why it matters to engineers

The cost of solving the wrong problem, and where the method is used

The cost of solving the wrong problem

- A technically perfect answer to the wrong question is worth nothing
- Errors made while framing the problem are the most expensive of all to correct – the cost of change rises steeply as a project advances
- Most products fail because nobody wanted them, not because the engineering was weak
- Design thinking moves the cheap learning to the front: talk to users and build rough prototypes before committing money, people and time

KEY POINT Prototype early and fail cheaply, or commit early and fail expensively. There is no third option.

Where design thinking is used

Products and hardware

Medical devices, appliances, vehicles, tools – anywhere a person has to hold, operate or maintain the thing you designed.

Services and experiences

Hospital admission, banking, campus registration – the design object is a sequence of steps a person has to get through.

Public systems and policy

Water supply, waste collection, public transport, disaster relief – many users, conflicting needs, no single customer.

Business and strategy

New business models, supply chains and organisational change, where the problem is defined by people's behaviour.

The mindsets of a design thinker

- **Human-centred** – begin and end with the people affected
- **Bias towards action** – a rough prototype teaches more than another meeting
- **Radical collaboration** – engineers, users and domain experts in one room
- **Comfort with ambiguity** – resist settling on an answer too early
- **Learn from failure** – a prototype that fails early has done its job
- **Show, do not tell** – make the idea visible so others can react to it

THINK ABOUT IT

Think of something you use every day that irritates you.

- What exactly goes wrong, and at which moment?
- Who else would have the same trouble – and who would not?
- Would a better product fix it, or does the real problem lie somewhere else?
- What would you need to find out before you designed anything at all?

What this course will ask of you

- **Unit I** – what design thinking is and why engineers need it
- **Unit II** – finding out what users actually need
- **Unit III** – generating and choosing between ideas
- **Unit IV** – building rough prototypes and testing them
- **Unit V** – sustainability, ethics and pitching your solution

ACTIVITY Form teams of four or five today. Your team carries one real problem through all five units and pitches it in the final session.

Summary

- A human-centred process and mindset for problems that arrive undefined
- Formalised between 1969 and 2008; now taught to engineers, not only designers
- Wicked problems must be discovered before they can be solved
- Framing errors cost the most, so the learning is moved to the front
- Used for products, services, public systems and strategy alike
- Six mindsets, beginning with human-centred and a bias towards action

Next: Engineering design vs traditional problem solving

- Well-structured and ill-structured problems
- Linear and iterative ways of working
- Divergent and convergent thinking; the Double Diamond
- Why designers reason from solutions, not only from problems